Sequence Labeling and Evaluations in Text Mining

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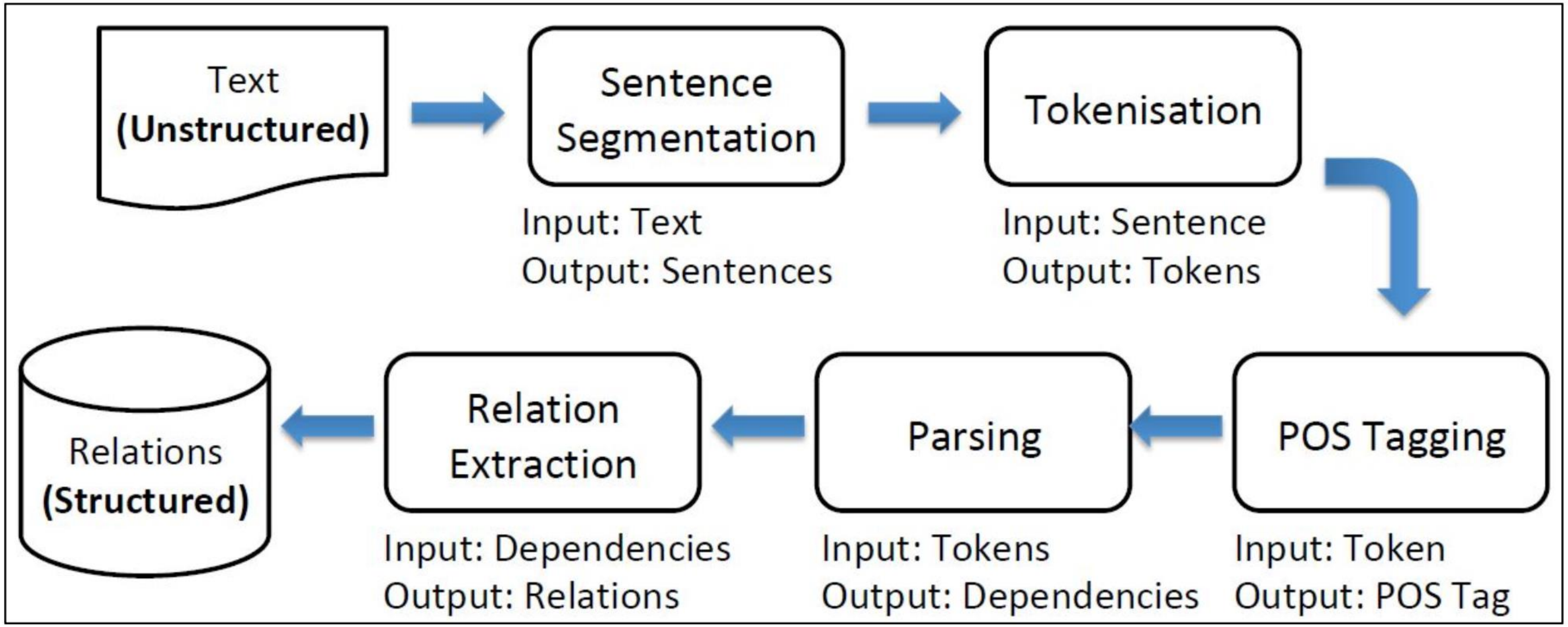
Module 1 **POS Tagging**

- What parts of speech are and why they matter
- Standard POS tagsets: **Penn Treebank** and **Universal POS**
- POS tagging as a **token-level classification task**
- Why POS tagging is difficult: **ambiguity and context**
- How context helps us choose the correct tag
- Rule-based vs. data-driven tagging approaches
- How tagged corpora provide supervision for POS models

The NLP Pipeline

- Natural language processing often works as a pipeline of components.
- Each component adds a layer of structure to raw text.



- Later components depend heavily on the quality of earlier ones.

What Are Parts of Speech?

- Parts of speech are classes of words defined by their grammatical role.

Nouns: *house*, *health*, *London*, etc...

Pronouns: *he*, *they*,…

Verbs: *walks*, *gave*, *showing*, …

Adjectives: *small*, *better*,…

Adverbs: *almost*, *happily*, …

Determiners: *the*, *a*, *an*

Conjunctions: *and*, *or*, *because*, …

Prepositions: *in*, *of*, *from*, …

- POS tags help us describe how words function in sentences.

Standard POS Tagsets

- POS tags come from standard annotation schemes.
- Two common tagsets are:
  - **Penn Treebank**: detailed, English-focused
  - **Universal POS tags**: broader, cross-linguistic
- Examples:
  - Penn Treebank: **NNP**, **VBD**, **DT**
  - Universal POS: **PROPN**, **VERB**, **DET**

CC Coordinating conjunction
CD Cardinal number
DT Determiner
EX Existential there
FW Foreign word
IN Preposition or subordinating conjunction
JJ Adjective
JJR Adjective, comparative
JJS Adjective, superlative
LS List item marker
MD Modal
NN Noun, singular or mass
NNS Noun, plural
NP Proper noun, singular
NPS Proper noun, plural
PDT Predeterminer
POS Possessive ending
PP Personal pronoun
PP$ Possessive pronoun
RB Adverb
RBR Adverb, comparative
RBS Adverb, superlative
RP Particle
SYM Symbol
TO to
UH Interjection
VB Verb, base form
VBD Verb, past tense
VBG Verb, gerund or present participle
VBN Verb, past participle
VBP Verb, non-3rd person singular present
VBZ Verb, 3rd person singular present
WDT Wh-determiner
WP Wh-pronoun
WP$ Possessive wh-pronoun
WRB Wh-adverb
Plus additional tags for punctuation

Penn Treebank
https://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html

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  - Universal POS: **PROPN**, **VERB**, **DET**

Open class words		Closed class words		Other	
ADJ	adjective	**ADP**	adposition	**PUNCT**	punctuation
ADV	adverb	**AUX**	auxiliary	**SYM**	symbol
INTJ	interjection	**CCONJ**	coordinating conjunction	**X**	other
NOUN	noun	**DET**	determiner		
PROPN	proper noun	**NUM**	numeral		
VERB	verb	**PART**	particle		
		PRON	pronoun		
		SCONJ	subordinating conjunction		

Universal POS tags
https://universaldependencies.org/u/pos/all.html

The POS Tagging Task

- The task is to assign one POS tag to each token in a sentence.
- POS tagging usually happens after tokenisation.
- Example:

*Book/**VB** that/**DT** flight/**NN** ./**.***

*Does/**VBZ** that/**DT** flight/**NN** serve/**VB** dinner/**NN** ?/**.***

- The same word can receive different tags in different contexts.

Why POS Tagging Is Hard

- Many words are ambiguous in isolation.

    Syntactic ambiguity (e.g., accidental homophones and homonyms)

    duck (action, verb)

    duck (bird, noun)

    Different syntactic roles

    To ***walk*** *vs to go for a* ***walk***

    Old *people vs the* ***old***

    They ***referee*** *the matches vs The* ***referee*** *starts the match*

Source: [*https://americanenglish.state.gov/*](https://americanenglish.state.gov/)

- In a lot of cases, POS tagging is therefore a **disambiguation** task.

How can we disambiguate?

Observe that syntactic ambiguity

... occurs when token is in **isolation**

... disappears when in combination with other words

*I want to **go** vs I want a **go***

*I can **walk** there vs I will take a **walk***

*The garbage **can** smell vs The garbage **can** smells*

... but sometimes it does not

*They **can** fish*

How can we disambiguate?

- A token is very unlikely to be a verb if its preceding word is a determiner
  I want a go
- A token is unlikely to be a noun if the immediately preceding word is *to*
  I want to go
- A token is more likely to be a possessive pronoun when followed by a common noun
  He stroked her cat
- ...but not always
  He gave her money

Context Resolves Ambiguity

- Local context often makes the correct tag much clearer.
- Examples:
  - *a walk* → **walk** is likely a noun
  - *to walk* → **walk** is likely a verb
  - *the old man* → **old** is an adjective
- POS tagging models learn how neighbouring words influence likely tags.

One Sentence, Two Valid Taggings

Meaning affects POS tagging.

- Example sentence: **I saw her duck.**
- Interpretation 1: I saw her **pet bird**.
  - *I/PRON saw/VERB her/PRON duck/NOUN ./PUNCT*
- Interpretation 2: I saw her **lower her head quickly**.
  - *I/PRON saw/VERB her/PRON duck/VERB ./PUNCT*
- A tagger must choose the tag that matches the intended meaning.

The Task

Assign POS tags to individual **tokens**

... but **only one** POS tag per token (for each run)

They can fish

Run 1: *They/**PRON** can/**AUX** fish/**VERB***

Run 2: *They/**PRON** can/**VERB** fish/**NOUN***

I saw her bat

Run 1: *I/**PRON** saw/**VERB** her/**PRON** bat/**NOUN***

Run 2: *I/**PRON** saw/**VERB** her/**PRON** bat/**VERB***

Rule-Based and Data-Driven Tagging

- A rule-based approach uses handcrafted grammatical rules.
- A statistical or neural approach learns patterns from annotated corpora.
- POS-tagged corpora provide:
  - example sentences
  - gold labels
  - evidence for disambiguation
- In practice, modern systems are usually data-driven.

Part-of-Speech Corpus: Penn Treebank example

Today	NN
is	VBZ
a	DT
nice	JJ
day	NN
.	.
I	PRP
want	VBP
to	TO
go	VB
for	IN
a	DT
walk	NN
.	.

Today/NN *is*/VBZ *a*/DT *nice*/JJ *day*/NN *.*/.

I/PRP *want*/VBP *to*/TO *go*/VB *for*/IN *a*/DT *walk*/NN *.*/.

(How to label this? Auto-regressive prediction. Use the previous token to provide context so to generate current token.)

Part-of-Speech Corpus: Universal Scheme example

Today	NOUN
is	VERB
a	DET
nice	ADJ
day	NOUN
.	PUNCT
I	PRON
want	VERB
to	ADP
go	VERB
for	ADP
a	DET
walk	NOUN
.	PUNCT

Today/NOUN *is*/VERB *a*/DET *nice*/ADJ *day*/NOUN *.*/PUNCT

I/PRON *want*/VERB *to*/ADP *go*/VERB *for*/ADP *a*/DET *walk*/NOUN *.*/PUNCT

Module 2 **Syntactic Parsing**

- What syntactic parsing is and why it matters
- How parsing goes beyond POS tags to represent **structure**
- **Dependency parsing**: heads, dependents, and grammatical relations
- How to read a dependency analysis
- Why parsing is needed for **attachment ambiguity**
- **Phrase structure parsing**: NP, VP, PP, and tree structure
- How dependency structure and phrase structure diffe

What Is Syntactic Parsing?

- POS tags tell us the category of each word.
- Parsing tells us **how the words are structurally related**.
- Parsing helps answer questions such as:
  - Which noun is the subject?
  - Which phrase is the object?
  - What does a prepositional phrase modify?
- Two major views of syntax are:
  - **dependency structure**
  - **phrase structure**

Why Parsing Matters

- Some sentences remain ambiguous even after POS tagging.
  - Example:
    - *The teacher discussed the essay with the student.*
  - Does *with the student* describe:
    - who the teacher talked to?
    - or which essay was discussed?
  - Parsing makes these structural choices explicit.

- Some sentences allow more than one syntactic analysis.
  - The same sequence of words can produce different structures and different meanings.
  - To understand a sentence, we cannot treat each word in isolation.
  - We need to determine what each word is attached or connected to.

Syntactic ambiguity

I saw the man on the hill with a telescope.

The violent police man injures the farmer with an ax.

Flying planes can be dangerous.

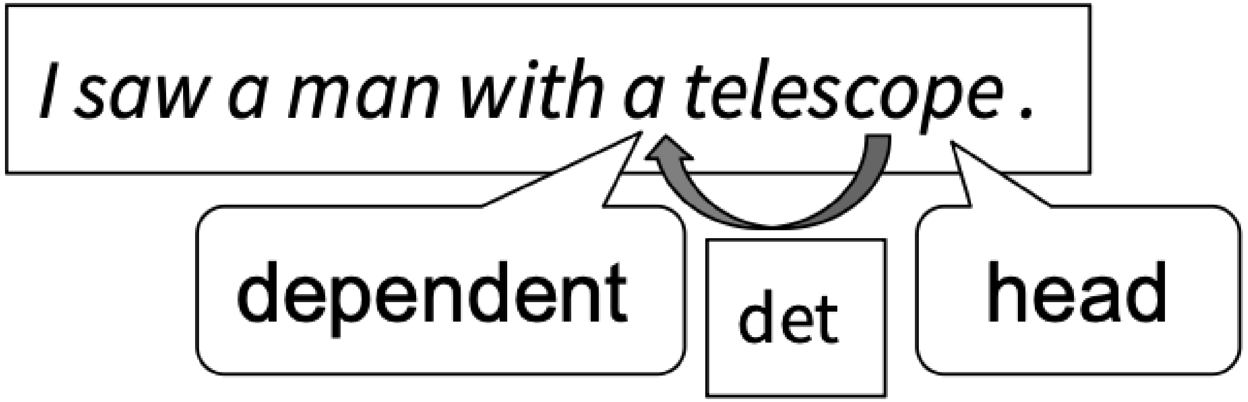
Visiting relatives can be boring.

In order to understand sentences, we cannot treat each word in isolation.

We need to determine what a word is **attached** or connected to.

Dependency Parsing: Core Idea

- A dependency parse represents grammatical relations between words.
- Each relation links:
  - a **head**
  - and a **dependent**
- The head of the whole sentence is usually the **main verb**.
- A dependency parse forms a directed structure over the tokens.

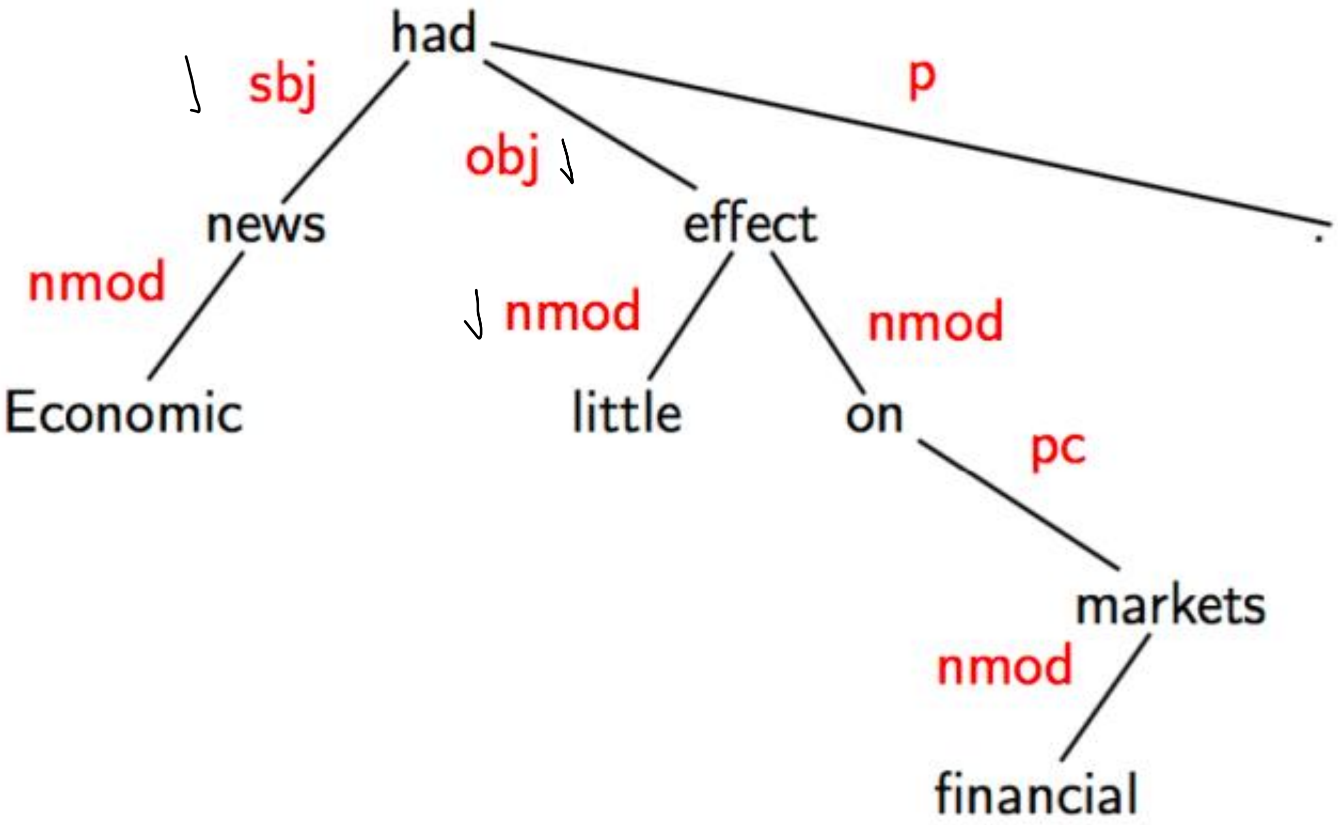


Heads, Dependents, and the Root

- In dependency parsing:
  - the sentence head is often the main predicate
  - subjects and objects depend on that predicate
  - modifiers depend on the word they modify
- Example:
  - In *The child opened the door*, **opened** is the root.
  - **child** is a subject dependent.
  - **door** is an object dependent.

A Dependency Tree in Practice

- In a dependency analysis, one token acts as the structural centre of the sentence.
- Other words connect to it through grammatical relations such as subject, object, and modifier.
- The labels on the edges tell us how each token functions in the sentence.
- This representation makes sentence structure explicit and easier to analyse



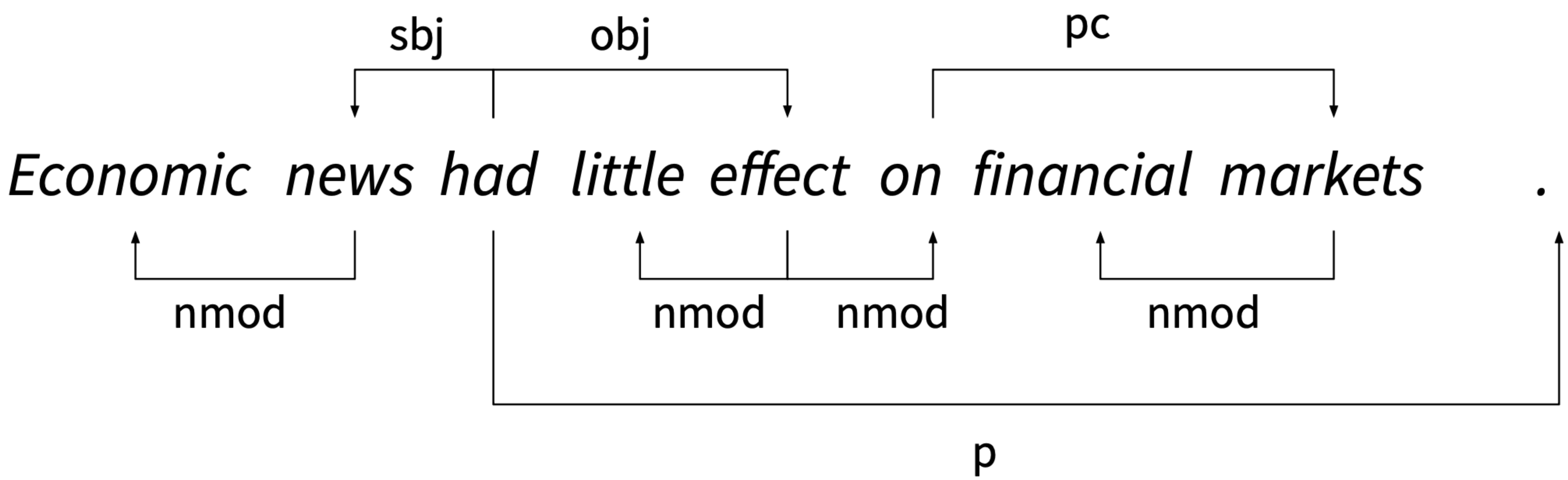
Common Dependency Relations

Label	Description	Example sentence	Example dependency
nmod/amod (nominal modifier)	phrase that modifies a noun phrase (NP)	Sam eats red meat.	nmod(meat, red)
aux (auxiliary)	non-main verb of a clause	He should leave.	aux(leave, should)
cc (coordination)	connects element of a conjunct and coordinating word	They either ski or snowboard.	cc(ski, or)
det (determiner)	between the head of an NP and its determiner	The man is here.	det(man, the)
obj/dobj (object)	between a verb and an NP denoting the entity acted upon	She gave me a raise.	obj(gave, raise)

Common Dependency Relations

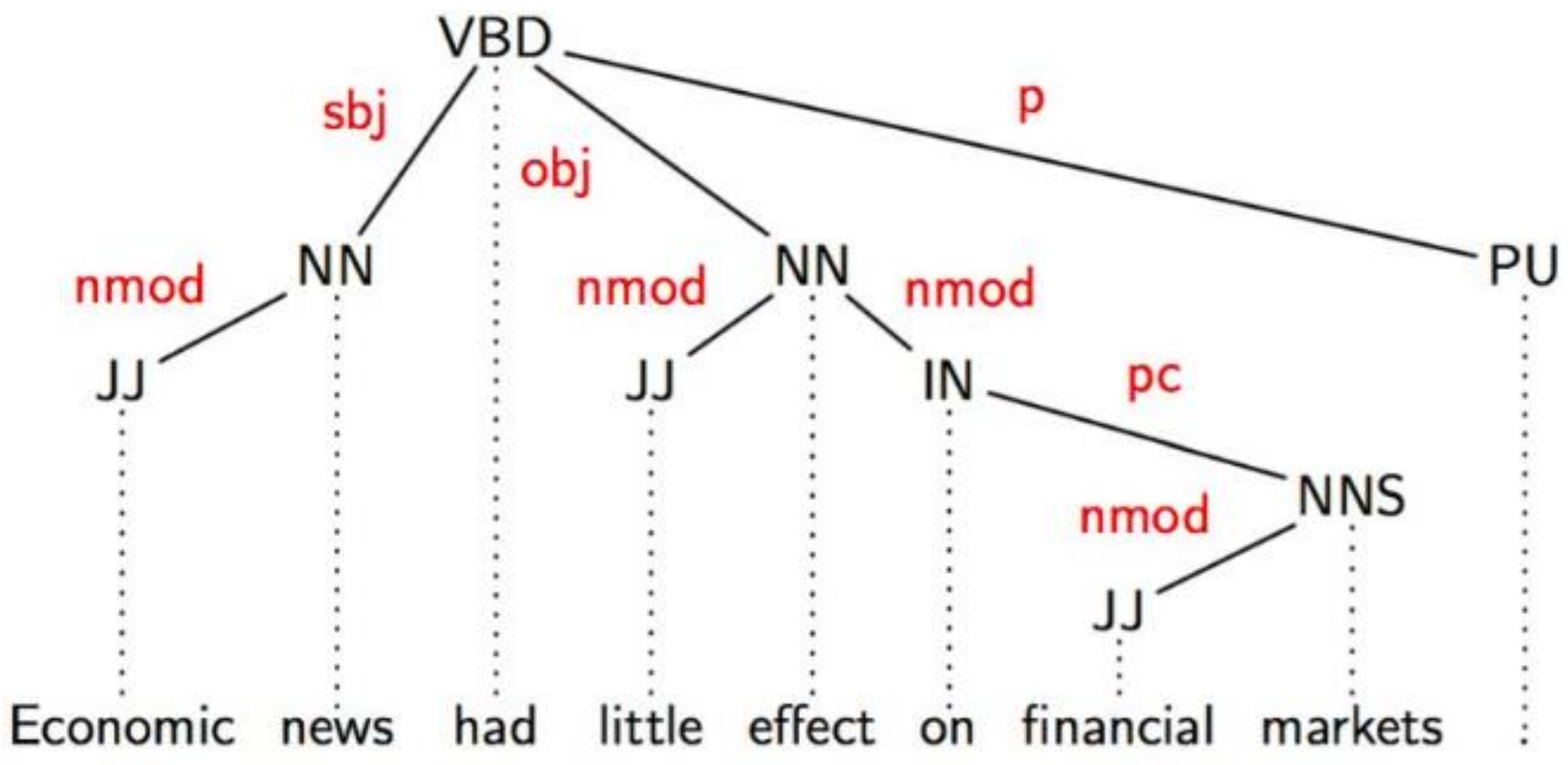
Label	Description	Example sentence	Example dependency
iobj (indirect object)	between a verb and an NP acting as indirect object	She gave me a raise.	iobj (gave, me)
vmod (verb modifier)	phrase that modifies a verb phrase (VP)	Genetically modified food.	vmod(modified, genetically)
sbj/nsubj (subject)	between a verb and an NP denoting the entity that serves as subject or agent	Clinton defeated Cole.	sbj(defeated, Clinton)
pobj/pc (object of preposition)	head of NP following a preposition	I sat on the chair.	pobj(on, chair)
p/punct (punctuation)	between the head of the sentence (the root) and punctuation	Go home!	punct(Go, !)

Example sentence



Reading a Dependency Analysis

- To read a dependency parse, ask:
  - What is the root?
  - Which words depend directly on the root?
  - Which words modify nouns?
  - Which phrases modify the event?
- This helps us identify the main structure of the sentence.



Attachment Ambiguity

(Parsing Tree)

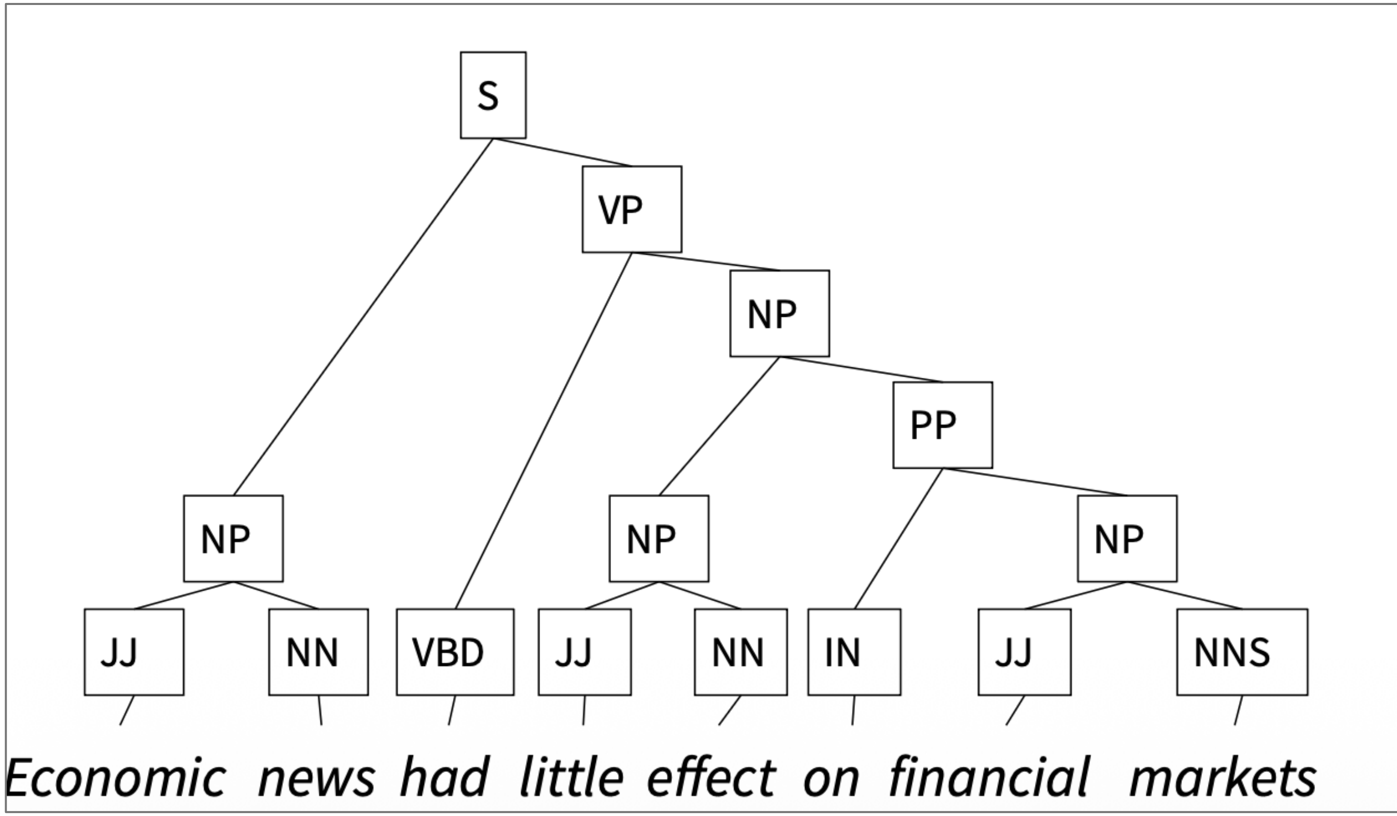
- A major challenge in parsing is **attachment ambiguity**.
- Example:
  - *The teacher discussed the essay with the student*.
- Interpretation A:
  - *with the student* modifies **discussed**
  - meaning: the teacher talked **to the student**
- Interpretation B:
  - *with the student* modifies **essay**
  - meaning: the essay is **about / associated with the student**
- Different meanings require different parses.

How to Decide Where a Phrase Attaches

- Ask: **Which word is this phrase modifying?**
- If the phrase describes the event or action, it usually attaches to the **verb**.
- If the phrase describes an entity, it usually attaches to the **noun**.
- Parsing is therefore closely tied to semantic interpretation.

Phrase Structure Parsing

- Phrase structure parsing represents sentences as **nested constituents**.
- Instead of head-dependent links, it groups words into phrases.
- Typical phrase types include:
  - **NP**: noun phrase
  - **VP**: verb phrase
  - **PP**: prepositional phrase
  - **AdjP**: adjectival phrase
  - **AdvP**: adverbial phrase



Building a Phrase Structure Tree

- Phrase structure analysis usually proceeds in stages:
  - identify POS categories
  - locate the major constituents
  - project phrase nodes
  - connect tokens to phrases
  - combine phrases into a full tree
- The result shows both grouping and hierarchy.

Analysis of Phrase Structure

Step 1: Label the syntactic category (POS) of each of the tokens

DT	NN	VBD	DT	NN	IN	DT	NN
The	*child*	*put*	*the*	*puppy*	*in*	*the*	*garden*

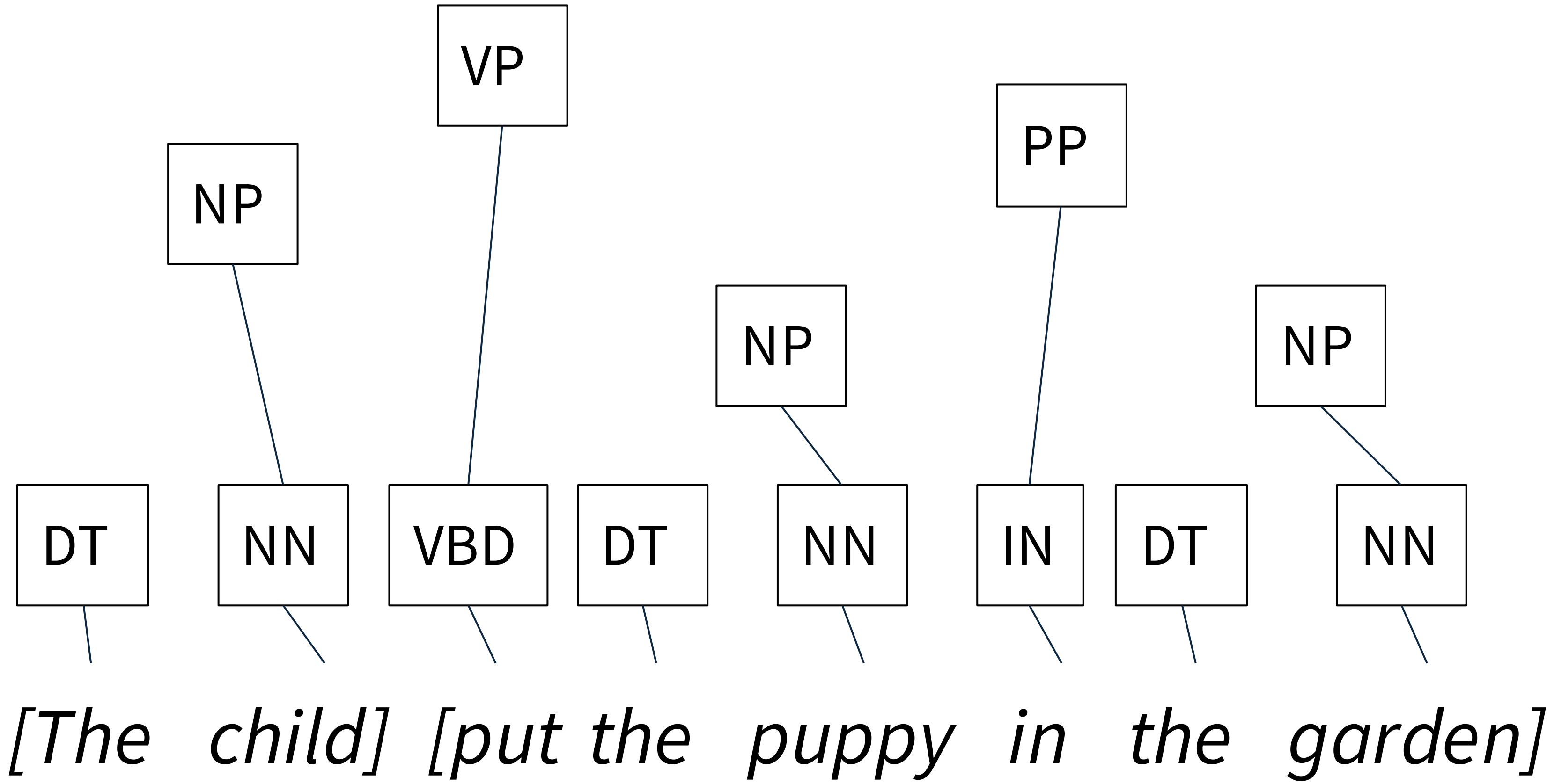
Analysis of Phrase Structure

Step 2: There are two principal constituents--a noun phrase (NP) and a verb phrase (VP). Locate the boundary between the two.

DT	NN	VBD	DT	NN	IN	DT	NN
[The	*child]*	*[put*	*the*	*puppy*	*in*	*the*	*garden]*

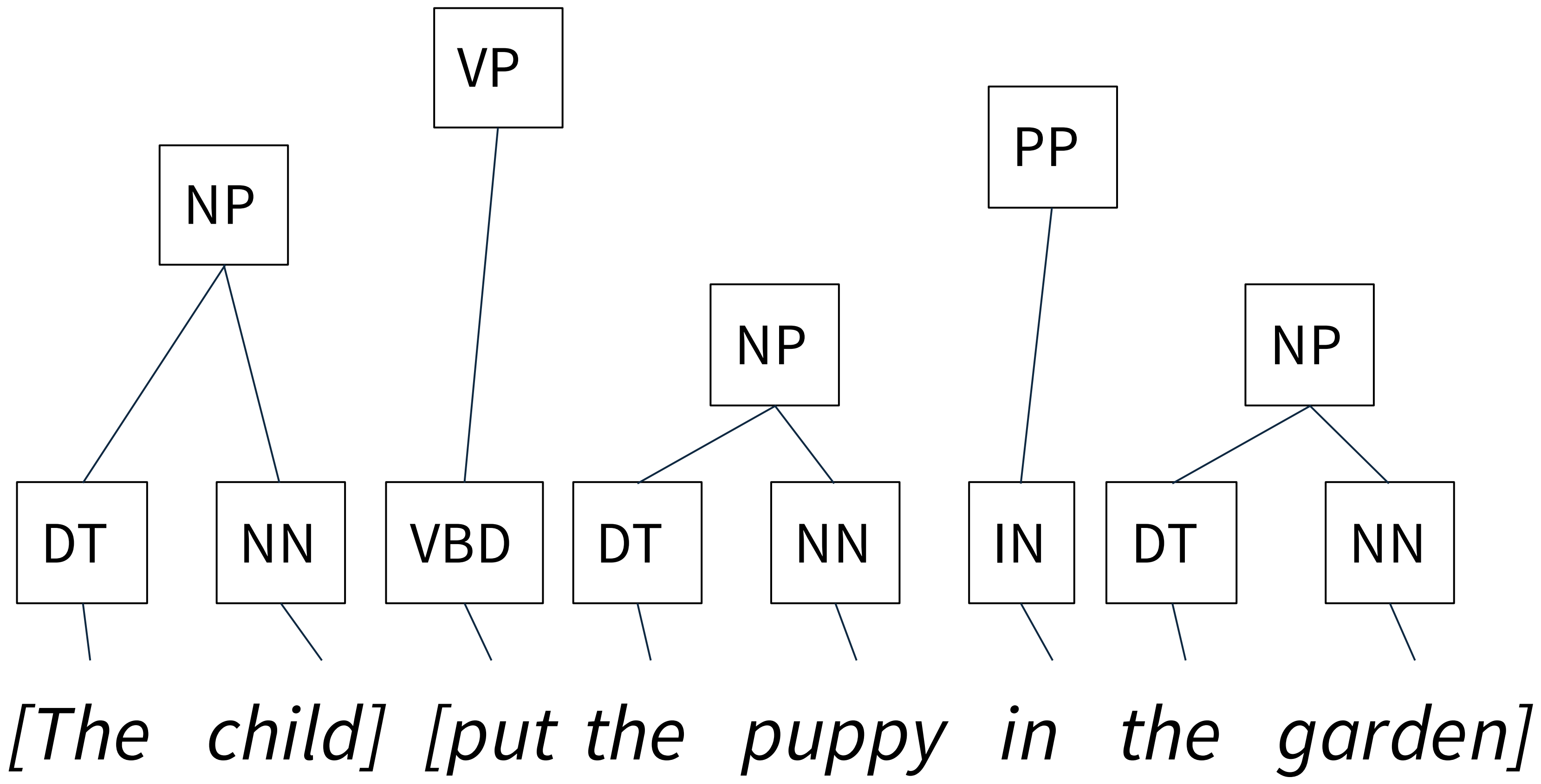
Analysis of Phrase Structure

Step 3: For each head noun/pronoun, verb, adjective, adverb and preposition, project a phrasal node: NP, VP, AdjP, AdvP, PP.



Analysis of Phrase Structure

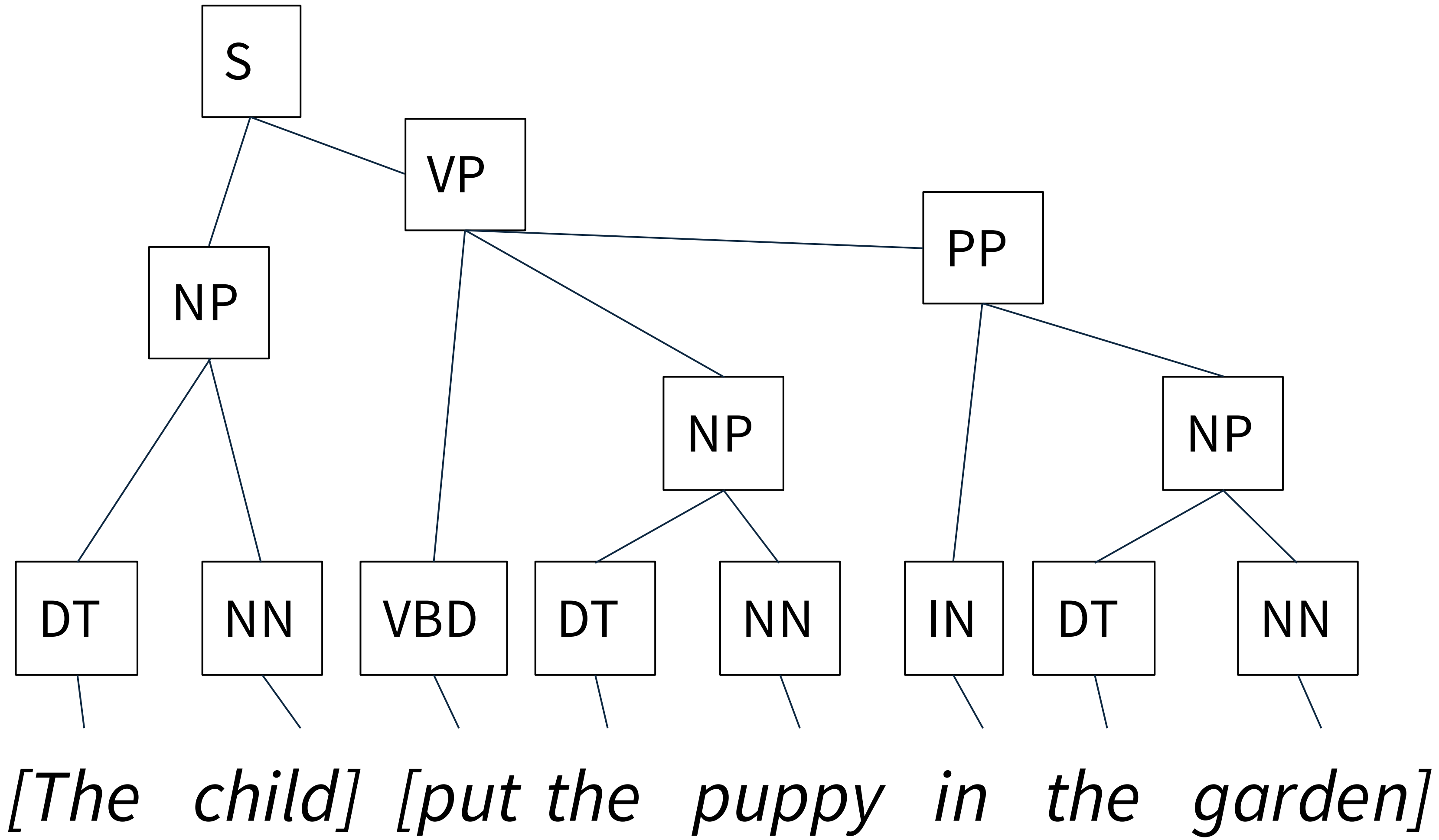
Step 4: Connect the remaining tokens to the nodes they belong to



Analysis of Phrase Structure

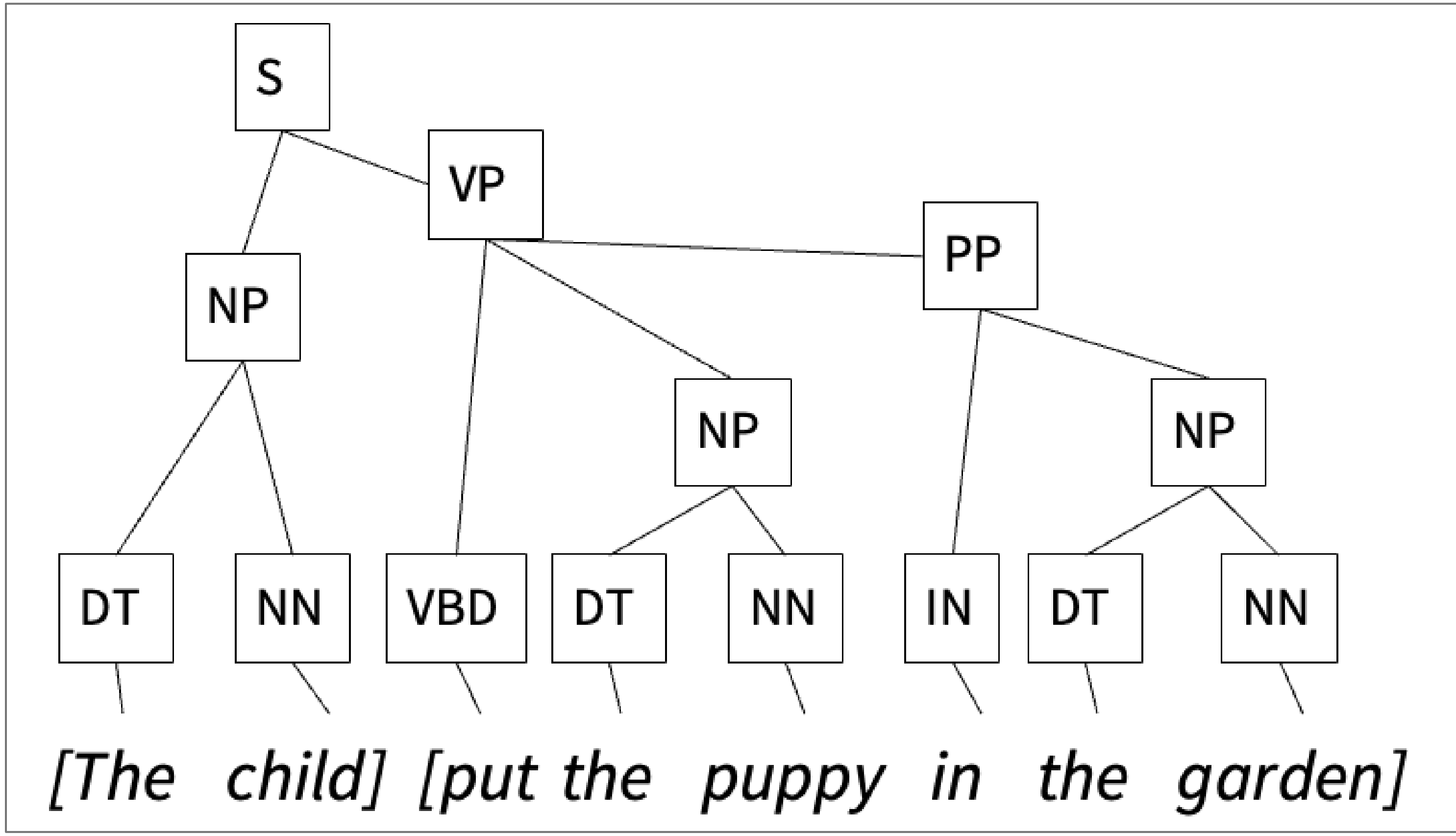
Dependency structure, task on exams.

Final tree



Identifying Noun Phrases

- A noun phrase is centered on a noun or pronoun.
- It may contain:
  - determiners
  - adjectives
  - possessives
  - modifiers
- Examples of noun phrases:
  - *the child*
  - *her old bicycle*
  - *the report on climate change*
- Not every long span in a sentence is a noun phrase.



Dependency Structure vs Phrase Structure

- **Dependency parsing** focuses on relations between words.
- **Phrase structure parsing** focuses on constituent groupings.
- Dependency parsing is especially useful for:
  - predicate-argument structure
  - information extraction
  - relation analysis
- Phrase structure parsing is especially useful for:
  - constituent identification
  - NP / VP / PP analysis
  - hierarchical sentence structure

Comparison: What can be represented?

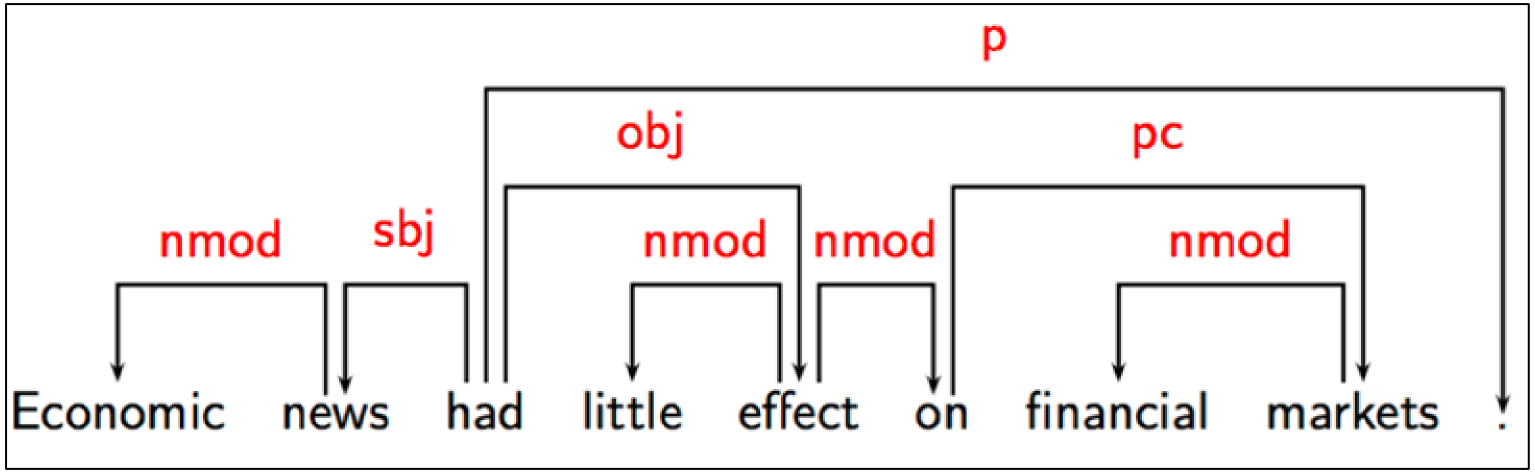
Dependency structure

- directed edges: head-dependent relations
- edge labels: grammatical functions
- POS tags: syntactic categories

Phrase structure

- non-terminal nodes: phrases (and their hierarchical structure)
- POS tags: syntactic categories

Dependency structures:
Endocentric vs Exocentric Constructions √



Construction	Head	Dependent	Required?
Exocentric	verb	subject (sbj)	yes
	verb	object(obj)	yes
Endocentric	verb	adverb (vmod)	no
	noun	adjective (nmod)	no

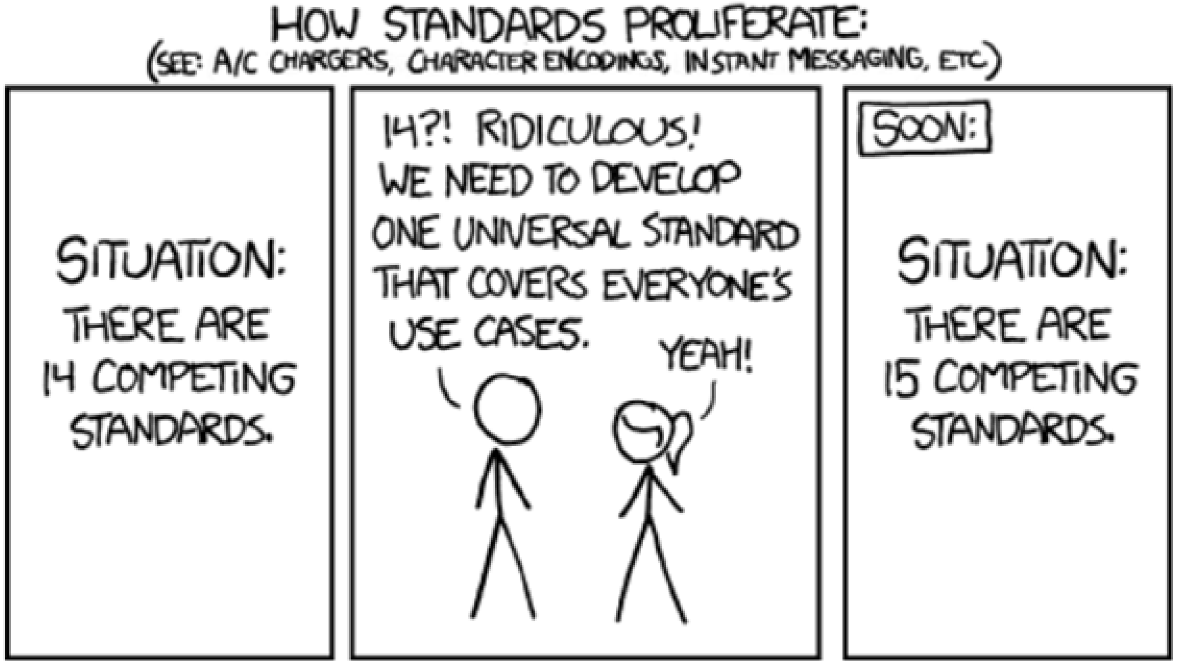
√

√

Endocentric is sometime optional.

Various notations

There are far too many!

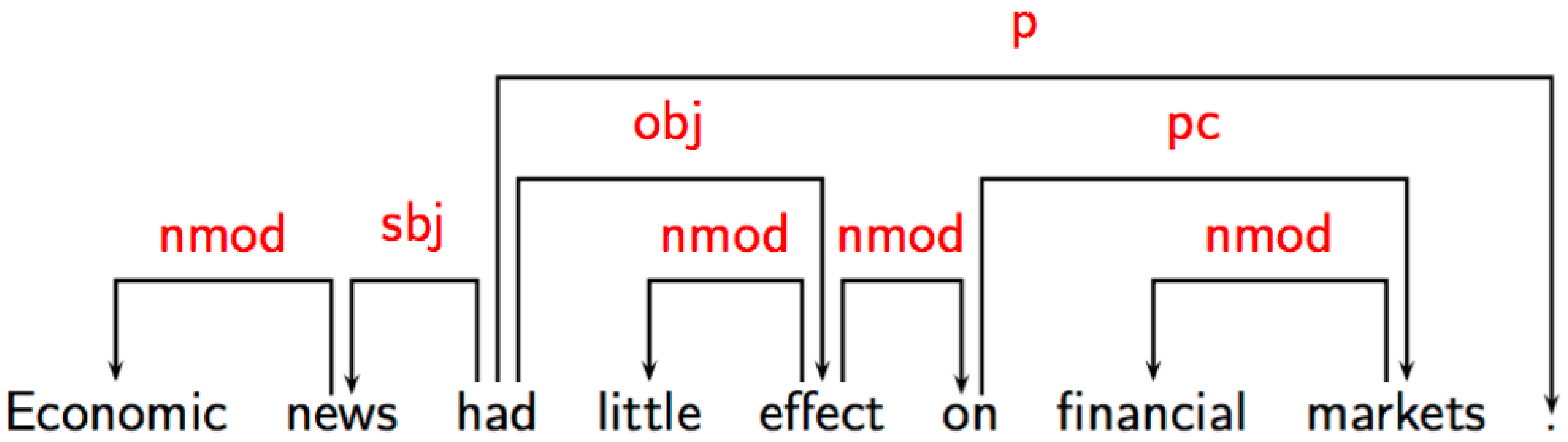


https://xkcd.com/927/

... But don't will
still see conceptual similarities.

Various notations

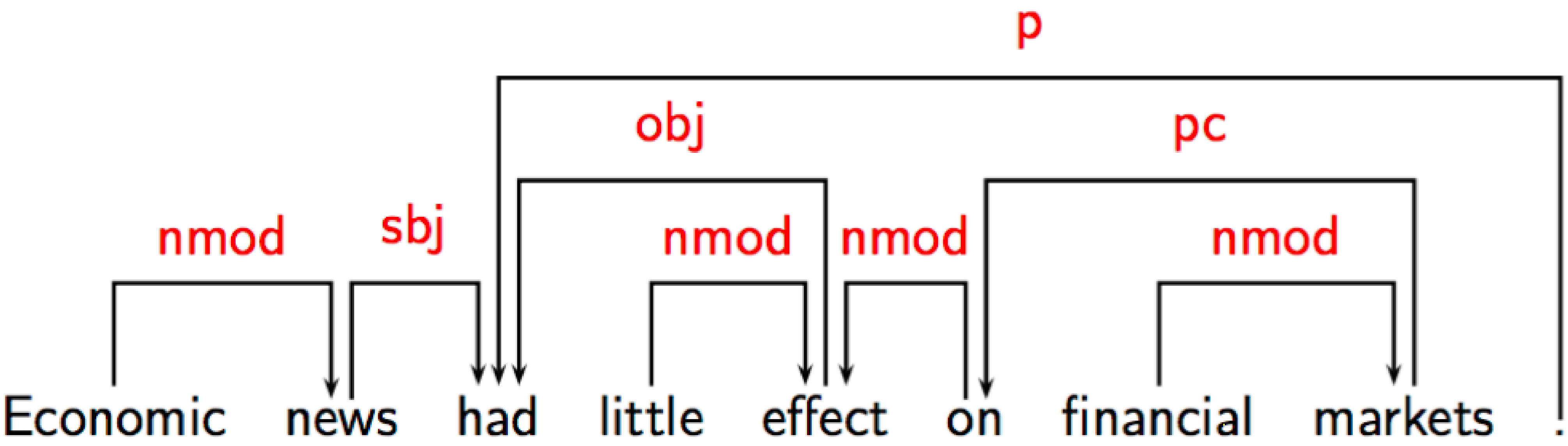
Notation 1: **Horizontally arranged tokens with directed edges**



Tokens + grammatical function

Various notations

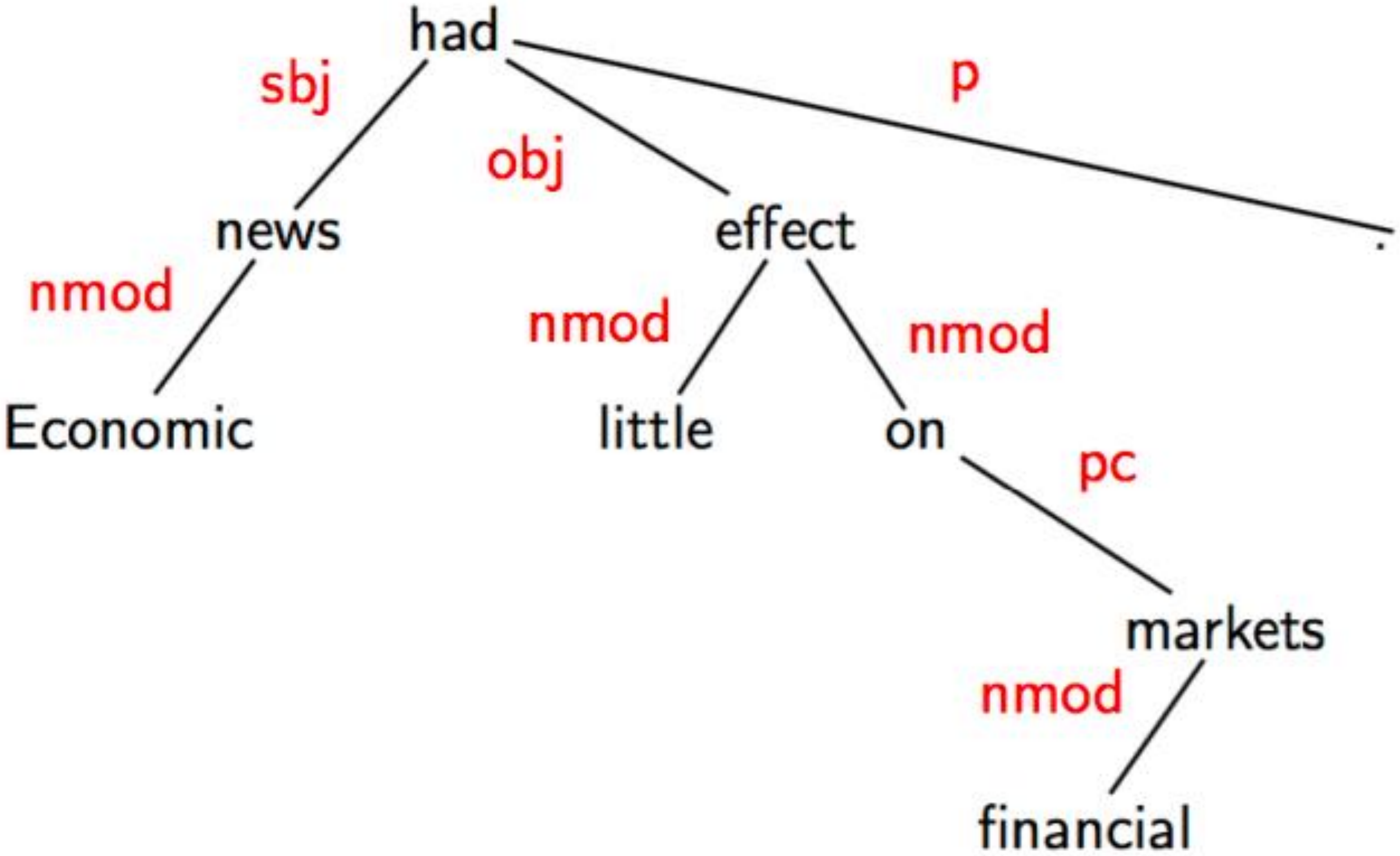
Notation 2: **Horizontally arranged tokens with directed edges**



Tokens + grammatical function

... but arrows are in the opposite direction!

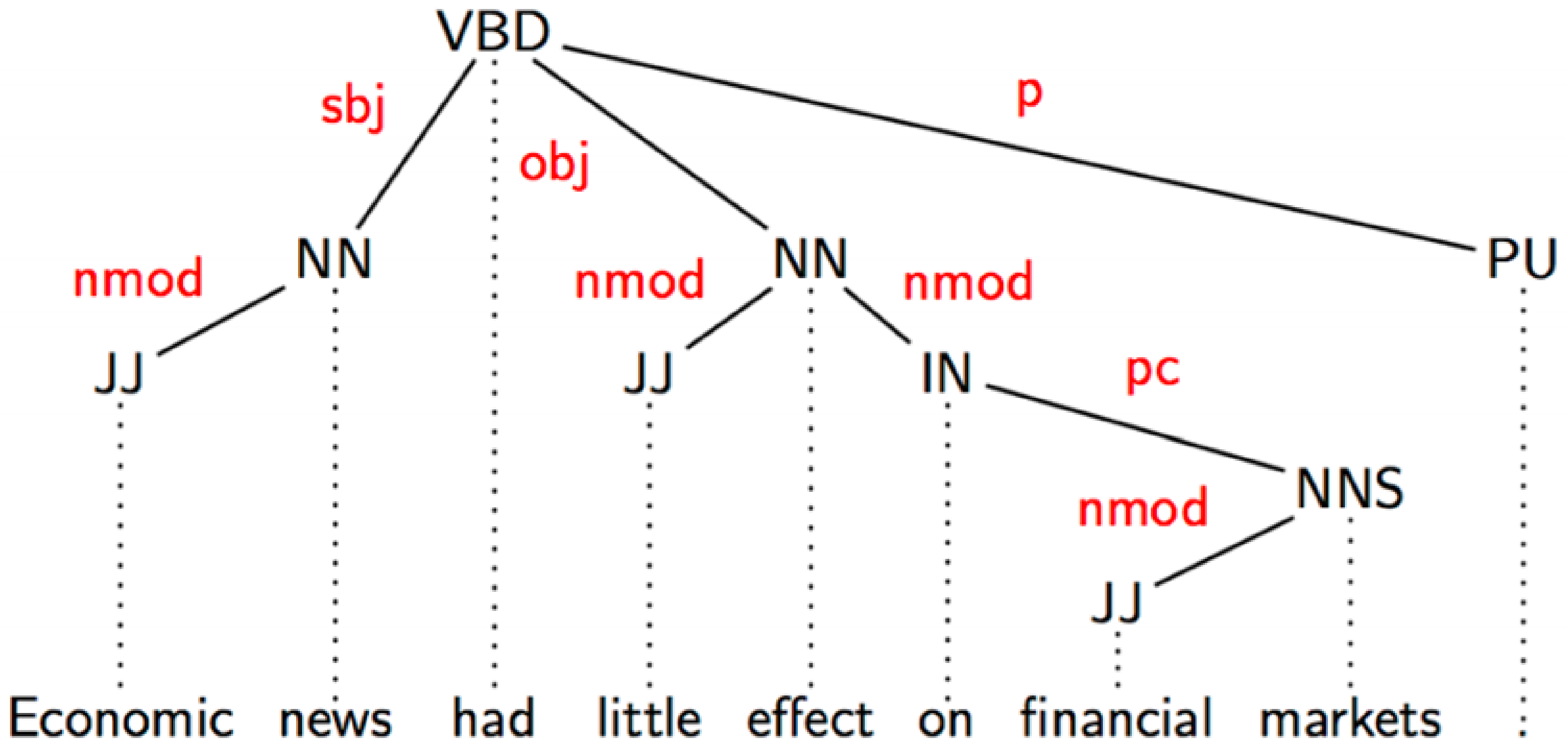
Various notations



Notation 3: **Tree of tokens**

Various notations

Notation 4: **Tree of POS tags mapped to tokens**



Tokens + grammatical function + **POS tags**

Module 3 **Named Entity Recognition**

Pos-tagging, syntatic parsing, name entity recognition

- What named entities are and why they are useful
- How entity types depend on the **task and domain**
- Why NER is **context-dependent**
- NER as a **sequence labelling** problem
- The **BIO tagging scheme** and entity spans
- How many output classes BIO requires
- Local vs. global sequence models
- CRFs, feature-based methods, and neural approaches
- Exact-match evaluation at the **entity level**

What Is Named Entity Recognition?

- Named Entity Recognition (NER) identifies mentions of real-world entities in text.
- It also assigns a label to each mention.
- Typical entity types include:
    - **Person**
    - **Organisation**
    - **Location**
    - **Date / Time**
    - **Money**
- NER helps answer questions such as:
    - Who is mentioned?
    - Where did something happen?
    - Which organisations are involved?

What is this snippet talking about?

Speaking at a Downing Street briefing on Monday evening, Mr Johnson added that the majority of people most in need of a vaccination in the UK might be able to get one by Easter.

And Prof Andrew Pollard - director of the Oxford vaccine group - said it had been "a very exciting day" and paid tribute to the 20,000 volunteers in the trials around the world, including more than 10,000 in the UK.

Location Time Person Organisation

https://www.bbc.co.uk/news/health-55040635

Named Entities Are Task-Dependent

- The useful entity types depend on the application domain.
- In news text, common types include:
    - person
    - organisation
    - location
    - date
- In biomedical text, useful types may include:
    - drug
    - disease
    - protein
    - virus
- NER is therefore not one fixed universal label set.

How about this?

Remdesivir (GS-5734), an inhibitor of the viral RNA-dependent, RNA polymerase with in vitro inhibitory activity against SARS-CoV-1 and the Middle East respiratory syndrome (MERS-CoV),5-8 was identified early as a promising therapeutic candidate for Covid-19 because of its ability to inhibit SARS-CoV-2 in vitro.[9]

Drug/Chemical Gene/Protein Virus Disease

Beigel, John H et al. "Remdesivir for the Treatment of Covid-19 - Final Report." The New England journal of medicine vol. 383,19 (2020): 1813-1826. doi:10.1056/NEJMoa2007764

Why NER Is Context-Dependent

- The same string can refer to different entity types in different contexts.
- Examples:
    - *Washington* may refer to a person, a city, or a government
    - *Amazon* may refer to a company or a river
- NER systems must use context, not just surface form.

NER as Sequence Labelling

- A common formulation treats NER as a token-level sequence labelling task.
- Each token receives a tag that tells us:
  - whether it is inside an entity
  - and which entity type it belongs to
- This makes NER similar to POS tagging, but with span boundaries added.

The BIO Tagging Scheme

- **B-X**: beginning of an entity of type X
- **I-X**: inside an entity of type X
- **O**: outside any entity
- Example:
  - *Maya Chen works at Northbridge Labs in Leeds*.
  - *B-PER I-PER O O B-ORG I-ORG O B-LOC O*
- BIO allows us to recover entity spans from token labels.

From BIO Tags to Entity Mentions

- Consecutive BIO tags define complete entity mentions.
- Example:
  - *B-PER I-PER* → one two-token person entity
- A correct entity prediction requires:
  - the correct **span**
  - and the correct **type**
- Token-level similarity is not enough if the entity boundary is wrong.

How Many Output Classes?

- If BIO is used for **N** entity types, the number of classes is:
    - **2N + 1**
- Why?
    - one **B** tag for each type
    - one **I** tag for each type
    - one **O** tag
- Example:
    - PER, ORG, LOC
    - classes = B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC, O
    - total = **7**

NER as a tagging problem (BIO scheme)

	Adam	Smith	works	for	IBM	in	London	.
POS tagging	NNP	NNP	VBZ	IN	NNP	IN	NNP	.
Entity recognition	B_PER	I_PER	O	O	B_ORG	O	B_LOC	O

***B**egin the mention*

***I**nside the mention*

- # classes = 2 * # entity types + 1

Local and Global Sequence Models

- A **local** model predicts each tag mainly from token-level context.
- A **global** model predicts a label sequence while considering dependencies between output tags.

  - Local approach: tags are ***independent*** each other
    - Any classifiers for sequence can be used, e.g., RNN, LSTM, BiLSTM
  - Global approach: tags are ***dependent*** each other
    - Hidden Markov Model (HMM)
    - Conditional Random Fields (CRF)

- Why does this matter?
  - some tag sequences are much more plausible than others
  - for example, an **I-ORG** tag should not normally appear after **O** without a preceding **B-ORG**

Conditional Random Fields

- A Conditional Random Field (CRF) is a global sequence model.
- It scores the entire output sequence rather than each token independently.
- CRFs are useful because they can model dependencies such as:
  - legal BIO transitions
  - consistency across neighbouring labels
- This often improves sequence labelling performance.

Linear-chain CRFs

Computation of probability

weight

feature function

$$p(y|x) = \frac{1}{Z_x}\exp(\Sigma_{t=1}^{T}\Sigma_{k=1}^{K} w_k f_k(y_t, y_{t-1}, x_t))$$

summation over all tokens

summation over all feature functions

Normalisation factor: to make sure the sum of probability is equal to 1

$$Z_x = \Sigma_y \exp(\Sigma_{t=1}^{T}\Sigma_{k=1}^{K} w_k f_k(y_t, y_{t-1}, x_t))$$

Features Used in Classical NER

- Classical NER models often use handcrafted features:
  - current word
  - surrounding words
  - orthographic features
  - prefixes and suffixes
  - trigger words
  - gazetteers
- Examples:
  - initial capitals
  - all digits
  - titles such as *Dr* or *Prof*
  - lists of country names or company names

This will be used to compute of probability of Linear-chains, finding out which of the word here is the most possible.

Feature types

- Contextual
  - current word wo
  - words around wo in [-3,…,+3] window
- Part-of-speech tag (when available)
- Trigger words
  - for person (Mr, Miss, Dr, PhD)
  - for location (city, street)
  - for organisation (Ltd., Co.)

Feature types (Cont.)

- Length (in terms of number of tokens)
- Orthographic (binary and not mutually exclusive)
  - initial-caps, all-caps, lonely-initial
  - all-digits contains-dots, punctuation-mark
  - single-char, contains-hyphen, URL
  - roman-numeral
- Suffixes (length 1 to 4)
  - each component of the NE
  - whole NE

Feature types (Cont.)

- Gazetteers
  - geographical locations
  - first names, surnames
  - company names
  - many others
  - whole NE is in gazetteer?
  - any component of the NE appears in gazetteer?

The more useful features you incorporate, the more powerful your learner gets!

Neural Approaches to NER

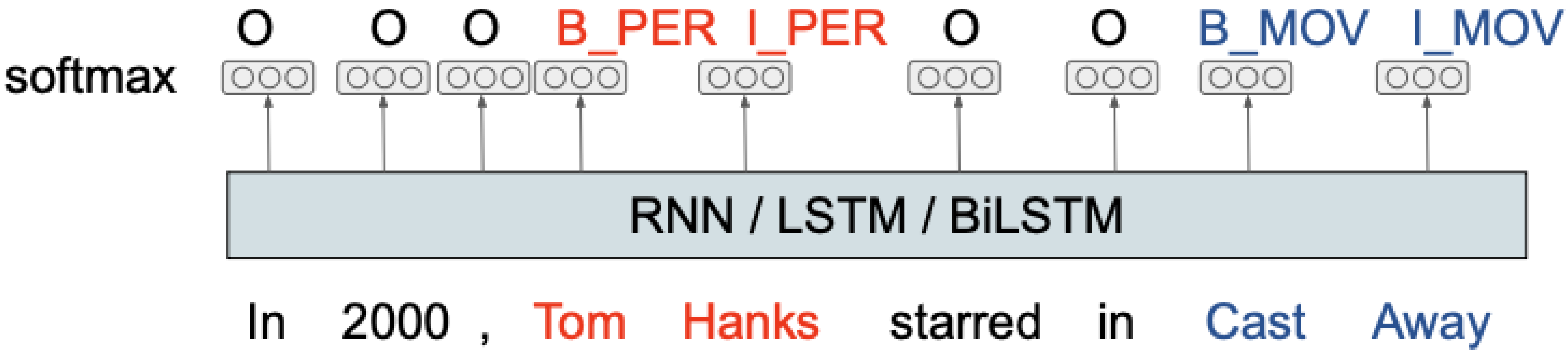
How does tokenization in transformer works?
How does Transformer-encoder learn word representation?

- Neural sequence models reduce the need for manual feature engineering.
- Common models include:
  - BiLSTM
  - BiLSTM-CRF
  - Transformer-based encoders
- These models learn contextual representations directly from data.
- They often perform well when training data is available.

Neural Approaches to NER

- Predicting tags independently

$$Pr(tag|token) = softmax(\mathbf{W}\mathbf{h}_{token} + \mathbf{b})$$



- Why do we use a recurrent net?

Exact-Match Evaluation for NER

- NER is usually evaluated at the **entity level**, not only at the token level.
- A predicted entity counts as a **true positive** only if:
  - its span is correct
  - and its type is correct
- If the boundary is wrong, it is not a correct entity.
- If the type is wrong, it is also not a correct entity.